

Shummit Proposal: Semantic AI Platform

Problem Statement

Many organizations report that a large share of their AI development effort is consumed by data preparation, integration, and aligning data across teams—not by developing models or intelligence. According to BCG research, organizations following the “10-20-70” principle spend about 10% of their effort on algorithms, 20% on technology & data, and 70% on people, process, and cultural transformation.

Teams across departments often build AI applications in isolation; they each solve recurring data connectivity, metadata, schema mismatch, or discovery challenges, and insights tend to stay within domain silos. The issue is not always lack of data, but lack of discoverability, shared vocabulary, or cross-team connection of data and semantic definitions.

Proposed Solution

We propose to build a Semantic AI Infrastructure Platform which enables teams to:

- Define shared business concepts once (e.g., *Person*, *Project*, *System*, *Event*) using web/semantic standards (e.g. JSON-LD contexts, RDF, etc.).
- Map their existing data to those shared concept definitions while keeping existing systems and workflows.
- Publish metadata/data mappings so that AI applications or analytic pipelines can discover and integrate relevant data across teams automatically without ad-hoc custom integration.

The idea is to shift some integration work outward: data-producing/publishing teams take responsibility for mapping; the system provides discoverability and linking.

Evidence & Academic Merit

Below are studies and observations that support technical feasibility, benefits, and known challenges; where precise metrics are unavailable, language has been generalized.

• Technical Feasibility & Case Studies

1. *MapSDI: A Scaled-up Semantic Data Integration Framework* (Jozashoori & Vidal, 2019) demonstrates that using mapping rules and preprocessing steps (projection, deduplication) guided by semantic mappings can reduce

knowledge graph creation time by an order of magnitude vs naïve approaches

2. *Knowledge Graph Induction enabling Recommending and Trend Analysis: A Corporate Research Community Use Case* (IBM Research) – shows how structured + textual data from different systems can be unified under a knowledge graph, and supports recommendation & analytics use cases
3. BCG’s “Where’s the Value in AI?” (2024) survey of many companies shows that only ~25-30% are moving beyond proofs of concept, with many still struggling to scale value; leaders in AI mature in both technical capability and organizational change
4. The “10-20-70” allocation model from BCG is repeatedly cited in recent AI strategy/consulting work, indicating that many organizations recognize that the people/process dimension is often the largest and hardest to address.

- **Benefits in Accuracy, Discovery, & Cross-Domain Use**

1. *A Benchmark to Understand the Role of Knowledge Graphs on LLM’s Accuracy for Question Answering on Enterprise SQL Databases* (Sequeda, Allemang, Jacob, et al., 2023) shows that for question answering tasks over an enterprise SQL schema, using a knowledge-graph representation + ontology + mappings increases accuracy substantially vs zero-shot LLM over raw SQL tables ($\approx 16\% \rightarrow \approx 54\%$).
2. *JEL: Applying End-to-End Neural Entity Linking in JPMorgan Chase* shows practical deployment of knowledge graph / entity linking in a large enterprise setting, with cross-system integration and use in risk/fraud/advice systems.

- **Industry / Organizational Observations & Challenges**

1. Many case studies (from e.g. Enterprise Knowledge) show organizations struggling with data discovery, inconsistent metadata, siloed systems, and decentralized content or practice, which semantic layers / knowledge graph / graph-based metadata systems help to address.
2. Industrial enterprise knowledge-graph development research (e.g. *Adoption of knowledge-graph best development practices* (Jawad et al., ScienceDirect, 2023)) documents that challenges include data quality, scalability of graph creation, governance, and tooling.

- **Proposal Limitations**

- There are very few publicly published large-scale ROI studies that directly compare, say, “semantic infrastructure + cross-team discoverability” vs “traditional ETL / integration approaches” in exactly similar settings.
- Many case studies are qualitative (improved insight, reduced duplicated effort) rather than fully quantified (dollar savings, latency improvements across the board).
- Performance metrics (for large/enterprise datasets) are often proprietary or not disclosed; so setting expectations must be cautious.

Technical Foundation

This doesn't have to be the way forward but platform may build on these tools and frameworks:

- JSON-LD (contexts, types) and RDF (including named graphs or similar concepts) are mature standards with broad adoption.
- Existing architectures / frameworks (e.g. mapping-rule frameworks like MapSDI) show that large heterogeneous sources can be integrated through semantic mappings.
- There are enterprise tools / graph databases (Neo4j, Stardog, GraphDB, Apache Jena, etc.) that support RDF/triples, graph querying, reasoning, etc. Many case studies use such tools.
- Performance / scale issues are tractable but non-trivial; approaches often require preprocessing, deduplication, incremental updates, good indexing, and governance to keep things maintainable.

Two-Week Deliverable

Week 1: Infrastructure Foundation

- Schema/ontology/context registry using JSON-LD/RDF standards
- Data / dataset registration API so teams can expose metadata / data mappings to the shared schema.
- Graph database instance with sample dataset(s) and ability to store named graphs or graph fragments with connections.

Week 2: Cross-Team AI Demonstration

- Simulating multiple organizational teams for a generic organization (i.e. HR, IT, Operations) mapping to shared concepts.
- Building a sample AI application or analytic dashboard that can use federated queries (SPARQL over multiple named graphs or datasets) to answer cross-domain questions that previously were difficult or impossible.
- Example queries: e.g., “How do system access patterns correlate with project outcomes across departments?”, “Are there patterns in employee performance, resource usage, and event logs that cross teams?” etc.

Success Metrics & Realistic Expectations

Technical / Performance Metrics

- Time to register a dataset with shared vocabulary (ideally low, but define realistic target depending on complexity)
- Discovery coverage: proportion of relevant cross-team datasets that are discoverable via the shared vocabulary / schema
- Query performance: federated or combined cross-team queries should run within acceptable latency (e.g. under some threshold, though in prototype settings may be higher)

Organizational / Adoption Metrics

- Stakeholder readiness and willingness to map to shared concepts
- Consistency / quality of mappings / metadata (data validation)
- Usability of tools (how easily teams discover data, how intuitive the mapping schema registry etc.)

Risk Mitigation & Organizational Readiness

- Known key challenges: governance, cultural resistance, lack of clarity in benefit, misalignment between domain teams, data quality, domain-specific terminology mismatches, etc.
- Mitigations:
 1. Start with willing, early adopter teams rather than trying to force whole org adoption.

2. Focus on compelling use cases that deliver visible value early (e.g. report/dashboard, cross-team insights) rather than architectures alone.
3. Secure executive sponsorship / leadership endorsement, to align incentives and resource allocation.
4. Build or adopt data governance frameworks / schema-management roles early (taxonomy/ontology stewards).

Strategic Impact & Scope

- The prototype aims to test whether adding more teams increases the “network effect” of semantic infrastructure: each new team that maps its data adds value for others.
- The goal is technical proof of concept and organizational value exploration, without overpromising large ROI until sufficient scale and maturity.
- Because many of the benefits are qualitative or indirect (reduced duplication, faster discovery, more consistent definitions, ability to reuse data/models), expectations should include both qualitative feedback and early quantitative indicators (e.g. speed ups in time to insight, reduction in repeated integration effort).

Sources

- Read these papers by Tony Seale first (there are several parts) gives a clear explanation of why an organization would want to adopt this approach to data management:
 - <https://medium.com/experience-stack/embrace-complexity-part-1-39483f10a47f>
 - https://medium.com/experience-stack/embrace-complexity-part-2-28eca3c94a8d?source=user_profile_page-----0-----2c658f4bef6d-----
 - https://medium.com/@Tonyseale/embrace-complexity-part-3-cloud-39f4074e4489?source=user_profile_page-----7-----2c658f4bef6d-----
 - <https://medium.com/@Tonyseale/building-your-connected-data-catalog-634674b41770>
- BCG – *Where’s the Value in AI?* (2024)
- BCG – AI Capabilities Overview (10-20-70 allocation)

- Jozashoori & Vidal (2019). *MapSDI: A Scaled-up Semantic Data Integration Framework*
- Sequeda, Allemang, Jacob, et al. (2023). *A Benchmark to Understand the Role of Knowledge Graphs on LLM's Accuracy for Question Answering on Enterprise SQL Databases*
- IBM Research (2022). *Knowledge Graph Induction enabling Recommending and Trend Analysis: A Corporate Research Community Use Case*
- JPMorgan Chase – *JEL: Applying End-to-End Neural Entity Linking* (2024)
- Enterprise Knowledge – Case Studies
- Jawad et al. (2023). *Adoption of knowledge-graph best development practices* (ScienceDirect)